

Treat the following exercises:

Exercise I: University

(3 pts)

In a university, there are different classrooms, offices and departments. A department has a name and it contains many offices.

A person working in the university has a unique ID and can be a professor or an employee.

A professor can be an essential, associate or assistant professor and he/she is enrolled in one department.

The offices and the classrooms have a number ID, and a classroom has a number of seats.

Draw the class diagram.

Exercise II: Travel agency

A customer arrives at a travel agency and wants to buy a ticket for a particular flight. The employee of the travel agency asks the customer the following information:

- The destination airport,
- The departure airport,
- The date and time of the departure,
- The class in which he wants to travel.

The employee enters this information into the flight information system which returns data about one or more possible flights matching the query. Each flight is described by:

- A number,
- A departure date,
- An exact time of departure,
- A time of arrival,
- A price.

The customer chooses one of the possible connections given by the system, reserves this flight and gets his ticket. The ticket contains more information such as the issue date and the amount paid.

The tickets received by the client must be paid. The airline agency supports two forms of payment; paying by credit card or by check. If the payment is made by credit card, the airline registers the number of the credit card and its expiration date. For the payment by check, the bank code and the number of the bank account are read from the check to perform an electronic debit note. Independent of the payment method, the airline agency always records the amount to be paid, the date of the payment and the signature of the customer.

1) Establish the corresponding use case diagram.

(3 pts)

2) Establish the conceptual class diagram from the use case described above.

(3 pts)

Exercise III: Fuel pump

User can buy fuel after inserting a credit card, which is read and validated by the controller.

If the card is invalid or the waiting time has been exceeded, the system returns to the initial waiting state.

If the card is valid, the user takes the hose out of the holster, and pushes the nozzle trigger to fuel his car.

When the nozzle is deactivated, the fuel flow is stopped and the price is charged on the credit card.

Draw the state diagram.

(3 pts)

Exercise IV: Voting

Once arrived at the voting station, the voter gives his own electoral card to the head of the voting station who checks if the voting station number is correct. Then, he checks the identity document and if it is valid he gives the ballot paper to the voter.

Then the head of the voting station waits for a cabin to get free and gives the pencil to the voter, a secretary signs the registry and

puts a print on the voter's certificate. Once the voter has voted, he/she inserts the ballot paper in the ballot box, gives back the pencil and takes back his/her own ID document.

Draw the activity diagram.

(3 pts)

Exercise V: Point of sale

The normal process of a point of sale in a supermarket is as follows:

- A customer arrives to the point of sale with his items to be paid.
- The cashier records the identification number of each item, as well as the quantity if it is greater than 1.
- The cash register displays the price of each item with its label.
- When all purchases are recorded, the cashier indicates the end of the sale.
- The cash register displays the total amount of the purchases.
- The cashier announces to the customer the total amount to be paid and asks him if he has a discount card. If the customer has a discount card he takes advantage of a 5% discount, then the cashier announces to the customer the new amount after the discount.
- The customer chooses his method of payment:
 - cash: the cashier takes the money, the cash register indicates the amount to be returned to the customer.
 - check: the cashier records the customer's identity number.
 - credit card: the demand of authorization is sent before the data entry.
- The cash register records the sale and prints it.
- The cashier gives the cash register ticket to the customer.

Design this situation by **sequence diagrams**, taking into account the alternative cases.

(3 pts)

Exercise VI: Stock management

In a store, a tradesman has a management system of the stock, whose functionalities are as follows:

- Edit the supplier's card.
- Possibility to add a new product (in this case, the supplier card is edited automatically. If the supplier doesn't exist, we can then create him).
- Edit the inventory: From this screen, the choice to print the inventory, to delete an article or to edit the product card is possible.

Model this situation by a use case diagram.

(2 pts)